

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

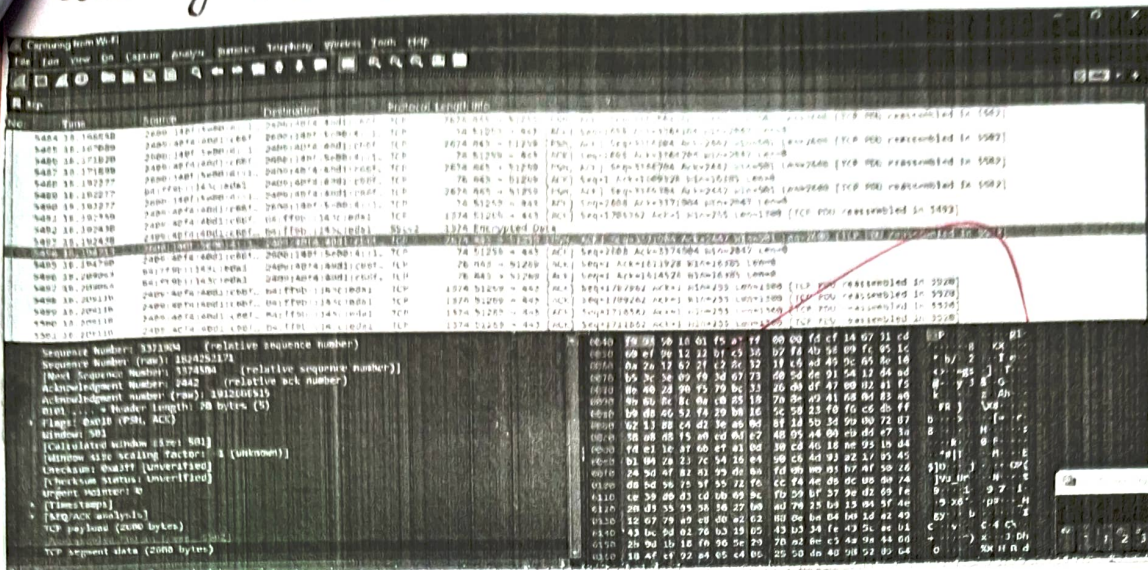
Assessment Tool Weightage: 40%

Code of Conduct:

I Deepak V-19241021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

Experiment 1: Capture TCP Three-Way Handshake using Wireshark



Three-way by the SYN, to establish

Materials Required:

- Computer
- Internet Connection
- Wireshark
- Web Browser

Procedure:

- 1) Open Wireshark and select active Network
- 2) Start Packet capture and open website
- 3) Access website to generate TCP communication
- 4) Stop packet capture after webpage loads
- 5) Identify Handshake packets after tcp filter

Observation:

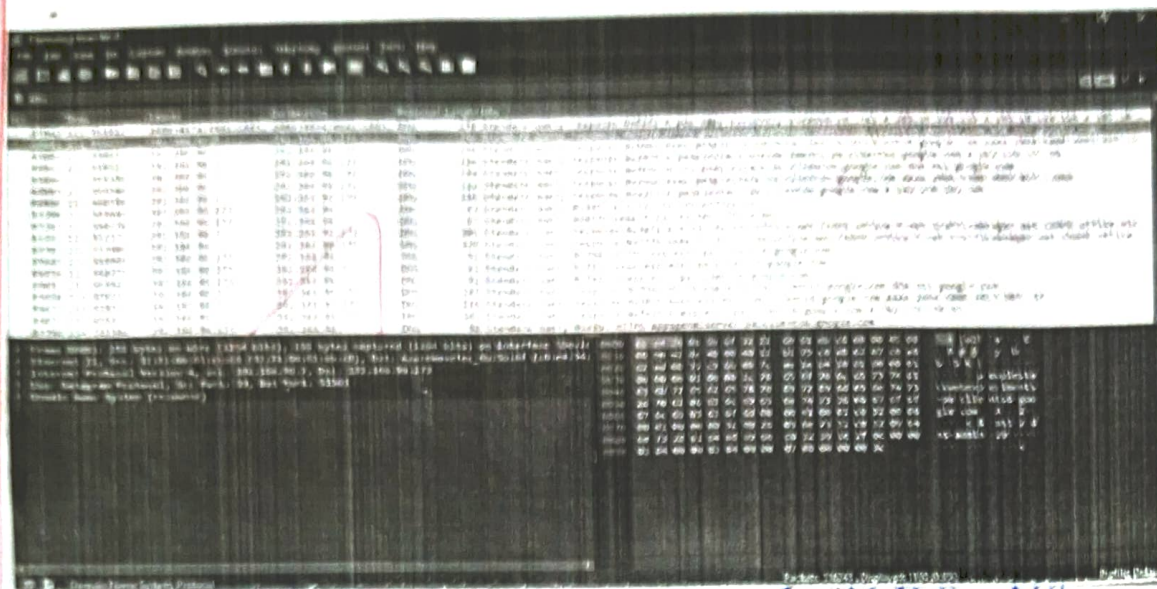
- SYN packet initiates the TCP connection
- SYN-ACK packet acknowledges the client's request
- ACK packet confirms successfully connection

Result:

The TCP Handshake was successfully captured

Experiment 2: Capture DNS Query using UDP

Wireshark



and compare

section

Web Browser

Procedure:

- 1) Open Wireshark and start capturing packets
- 2) Visit website to generate DNS query traffic
- 3) Stop capture after successful webpage loading completes.
- 4) Apply dns filter to display DNS packets.

Observation:

- DNS queries use UDP on port 53.
- UDP header size is only 8 bytes
- TCP header size is atleast 20 bytes
- UDP provides faster but less reliable communication

Result:

The DNS query was captured successfully, demonstrating UDP's light-weight communication.

A.



Connection

slr

- 1) Open Wireshark and begin capturing packets
- 2) Generate traffic by browsing websites.
- 3) Disconnect network $\&$ briefly to simulate packet loss
- 4) Reconnect and compare TCP with UDP

- TCP retransmitted. and begin capturing network packets. lost packets.
- UDP transmitted packets without Retransmission

TCP retransmitted lost packets, while UDP did not.

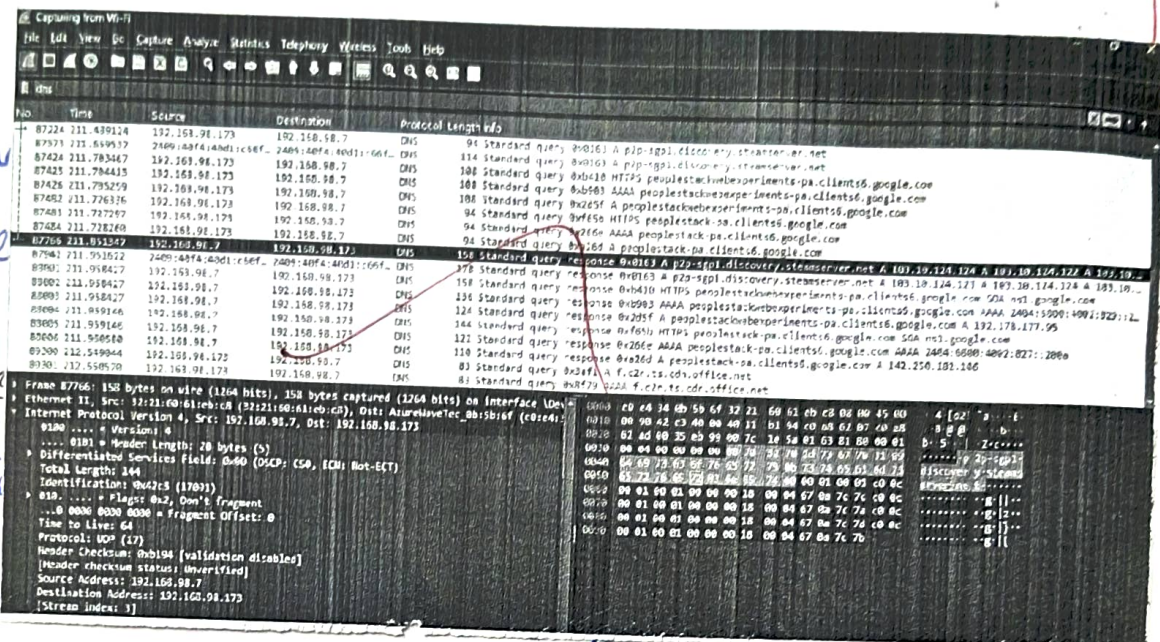
Experiment 4: Capture DNS Traffic Using Wireshark

Aim:

To capture query and response packets

Apparatus Required:

- Computer
- Wireshark



Procedure:

- 1) Open Wireshark and start capturing packets
- 2) Visit website to generate DNS query traffic
- 3) Stop capture after successful webpage loading
- 4) Apply dns filter to display DNS packets

Observation:

- DNS query type identified as A CTPv4 Address.
- DNS response contained the requested IP address.
- Resolution time was approximately 10-20 ms.
- DNS used UDP for faster resolution

Result:

The DNS query and response were captured successfully using UDP communication.